

RITHU SAJI | Senior AI and Automation Engineer | Immediate Joiner

rithusaji99@gmail.com | 79079 85131 | Kochi, Kerala | www.linkedin.com/in/rithu-saji-9b1b36198

PROFILE

AI & Automation Engineer with around 5 years' experience building AI-driven and RPA-powered solutions across healthcare, finance, and enterprise automation. Experienced in deploying production-grade AI systems integrating ML, NLP, and automation tools.

EDUCATION

Degree	Institution	Duration	Score
MTech, Artificial Intelligence and Machine Learning <i>WILP</i>	<i>Birla Institute of Technology and Science, Pilani</i>	2023 –2025	CGPA: 8.31
BTech (Honours), Electrical and Electronics Engineering <i>APJ Abdul Kalam Technological University</i>	<i>Mar Athanasius College of Engineering</i>	2017 – 2021	CGPA: 8.75
Class 12 CBSE	<i>Rajagiri Public School, Kalamassery</i>	2016 - 2017	94.6%
Class 10 CBSE	<i>Rajagiri Public School, Kalamassery</i>	2014 - 2015	CGPA:10

WORK EXPERIENCE

Senior AI and Automation Engineer

Digital Worker Services (Kloudpad Mobility Research)

Dec 2024 – Present

- **Candidate Screening System for NHS UK** [*Automation Anywhere Professional Services*]
 - Designed an **AI-based Candidate Screening System** for NHS UK using Automation Anywhere Knowledge API (RAG model), reducing screening time across departments.
 - Received **appreciation from Automation Anywhere Professional Services** for the project.
- **Tax Account Roll Forward Automation** [*Automation Anywhere Centre of Excellence (COE)*]
 - Developed an **automation system** to streamline the **Tax Account Roll Forward (TARF)** process using **Workday data**.
 - Automated **data validation, and reconciliation** for quarterly tax reporting.
- **Financial Chatbot** -For querying **financial statements** and **company performance metrics**.

Tech Stack: Python, PyTorch, Hugging Face, scikit-learn, ChromaDB -vector database, Streamlit

- Implemented **Retrieval-Augmented Generation (RAG)** for accurate, context-aware responses.
- Used **TF-IDF + Logistic Regression** for intent classification to route domain-specific queries.

- Generated **semantic embeddings** with **Hugging Face all-MiniLM-L6-v2** and stored/retrieved data in **ChromaDB** using cosine similarity search.
- Integrated a **Small Language Model (microsoft/phi-3.5-mini)** via PyTorch and Hugging Face for response generation.
- Built an **interactive Streamlit interface** with real-time query handling, context retrieval visualization, and conversational responses.

Senior Software Engineer

Digital Worker Services (Kloudpad Mobility Research)

Nov 2023 – Dec 2024

- **DataDoc- Intelligent SaaS** for Document Data Extraction
 - Developed a Flask-based SaaS for automated data extraction from bulk PDFs using **OpenAI GPT-4o** and **Azure Document Intelligence**.
 - Built a web interface for multi-file upload, custom field selection, and Excel export.
 - Deployed on Microsoft Azure, enabling secure, scalable document processing with automated email delivery.
- **SolutionSeeker- an AI-based ticket resolution** assistant.
 - Integrated with **ManageEngine API** to retrieve and update ticket details.
 - Used **OpenAI API** to generate contextual ticket resolutions for technician review.
 - Improved support efficiency by **reducing ticket response and resolution time**.

Systems Engineer

Tata Consultancy Services – Nielsen IQ (EBIT & NIV Projects)

Jul 2021 – Nov 2023

- Developed **PySpark ETL pipelines** on **Azure Databricks** for large-scale data processing and transformation.
 - Automated workflows using **Apache Airflow**
 - Implemented **CI/CD pipelines** with **Jenkins** and **Bitbucket**
- Tech Stack:** PySpark, Azure Databricks, Airflow, Swagger API, Jenkins, Bitbucket, WinSCP, Python

SKILLS

Category	Skills
AI & Machine Learning	• Machine Learning, Deep Learning (CNNs, RNNs, Transformers) • NLP & Conversational AI – LLMs, RAG, Embeddings, Vector Databases (ChromaDB, FAISS), LangChain, NLTK, SpaCy • Large Language Models – OpenAI GPT-4o, GPT-4-Vision, GPT-3.5-Turbo • Small Language Models – Microsoft Phi-3.5-mini, all-MiniLM-L6-v2 • Pre-trained Models & Libraries – Hugging Face Transformers, Ollama • Enterprise AI Platforms – Automation Anywhere Enterprise Knowledge (RAG Model) • AI Services – Azure Document Intelligence • Development Tools – Jupyter Notebook, Google Colab

	• Frameworks – PyTorch, TensorFlow, Keras, Scikit-learn
AI powered tools /IDE	Visual Studio code-GitHub copilot, Claude, Gemini, OpenAI GPT models for debugging
Data (ETL Pipelines)	• Azure Databricks, Apache Airflow, PySpark, SQL, Data Preprocessing & Transformation
MLOps & Deployment	• Docker, MLflow, Azure App Service • CI/CD Pipelines (GitHub Actions, Jenkins, Bitbucket) • Model Deployment
API & Web Frameworks	• Flask, Django
RPA & Intelligent Automation	• Automation Anywhere (Enterprise Knowledge, AA Copilot, Document Automation, UI Automation) • Power Automate

CERTIFICATIONS

- [Automation Anywhere Certified Master RPA Professional \(Automation 360\) – Automation Anywhere \(2024\)](#)
- [Automation Anywhere Certified Advanced RPA Professional – Automation Anywhere](#)
- [MATLAB Certified – MathWorks](#)
- [Pixiebrix Professional Certification \(Embedded Automation\)](#)

OTHER PROJECTS

1.MTech Dissertation: SmartKare: An AI-Driven Django Application for comprehensive Healthcare Monitoring

Tech Stack: Python, Django, Scikit-learn, Matplotlib, SQLite, Azure

Algorithms: Isolation Forest, Random Forest, Z-Score

- Built an **AI-powered healthcare monitoring system** for real-time patient vital analysis and anomaly detection.
- Implemented **Isolation Forest and Z-Score algorithms** to detect abnormal patterns in vitals such as heart rate, SpO₂, and temperature.
- Developed **dashboards** in Django using **Matplotlib** for health trend visualization.
- Integrated **automated alert notifications** for caregivers/immediate family upon anomaly detection.
- Used **role-based access control (RBAC)** for secure multi-user data management (patients, caregivers, and experts).
- **Optimized model parameters** to improve anomaly detection accuracy and reliability.
- **Deployed the end-to-end Django system on Microsoft Azure**

2.IR-Based Factoid Question Answering System (Flask, NLP, TF-IDF, BM25)

- Built a **Flask-based QA system** that retrieves factual answers from PDFs using **TF-IDF** and **BM25** information retrieval techniques.

- Applied **NLP (NLTK)** for question classification and passage extraction with a simple **web interface** for user interaction.

3.AI powered Alzheimer's Detection from MRI Scans

- Developed a **CNN classifier** for Alzheimer's stages (Non-Demented, Very Mild, Mild, Moderate Demented).
- Applied **SMOTETomek balancing, pixel normalization**, and evaluated with **precision, recall, F1, AUC, and ROC curves**.

4.AI-Powered Social Distancing Detection

- Implemented a **YOLO-based computer vision system** to detect and alert social distancing violations in real time during COVID-19. ([Published in International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering \(IJAREEIE\)-DOI:10.15662/IJAREEIE.2021.1006026](#))

INTERESTS

Writing – Copyright Registered (2023): English Poetry Collection on human emotions & psychology

REFERENCES

Available upon request